

Are You Getting What You Pay For?

Auditing Model Substitution in LLM APIs: Why Software Tests Fail and Hardware Attestation Wins

Commercial LLM APIs create a verification gap: users pay for named models, while black-box serving prevents direct proof that the advertised weights ran.

arxiv.org/abs/2504.04715

technical audit briefing

Trust gap

Advertised
model



Black-box
API



Actual
inference

The Model Substitution Problem: Economic Incentives Drive Covert Swaps

Economic incentive

Named commercial APIs charge for premium models, but providers can reduce GPU costs by serving cheaper alternatives while maintaining pricing.

Formal audit question

$H_0: P_{\text{actual}}(y|x) = P_{\text{spec}}(y|x)$

$H_1: P_{\text{actual}}(y|x) \neq P_{\text{spec}}(y|x)$

black-box API boundary



01 Smaller model

8B instead of 70B

02 Quantized variant

FP8 / INT8 instead of FP16

03 Fine-tuned version

updated or adapted weights

04 Different family

entirely different architecture

Detection is a distributional test over API samples—not direct inspection of weights or execution.

FP8 Quantization Preserves Most Accuracy — Making Substitution Hard to Detect

Model	MMLU		GSM8K		MATH		GPQA Diamond	
	Orig.	FP8	Orig.	FP8	Orig.	FP8	Orig.	FP8
Llama-3-8B-Instruct	62.69 ± 0.18	62.43 ± 0.26	61.14 ± 3.47	60.90 ± 4.10	20.65 ± 3.43	14.91 ± 2.49	22.62 ± 0.22	20.14 ± 0.24
Llama-3-70B-Instruct	78.05 ± 0.08	77.88 ± 0.13	88.06 ± 1.44	87.35 ± 1.24	35.69 ± 1.33	35.75 ± 1.16	29.60 ± 0.51	33.12 ± 0.30
Gemma-2-9b-it	71.86 ± 0.08	71.92 ± 0.11	81.80 ± 1.35	79.41 ± 1.14	33.41 ± 0.28	32.53 ± 0.34	28.64 ± 2.97	27.81 ± 3.22
Qwen2-72B-Instruct	82.18 ± 0.08	81.98 ± 0.08	86.72 ± 1.00	86.82 ± 0.97	37.39 ± 1.41	37.67 ± 1.39	29.93 ± 2.71	31.08 ± 1.96
Mistral-7B-Instruct-v0.3	59.15 ± 0.10	58.77 ± 0.13	35.90 ± 4.54	32.20 ± 4.02	8.94 ± 1.24	7.68 ± 1.23	21.60 ± 0.17	22.72 ± 0.19

Pattern: FP8 deltas are small on MMLU and GSM8K and often within reported error bars, so black-box benchmark checks have weak power against quantization substitution.

Higher Temperature Amplifies Performance Gaps but Increases Noise

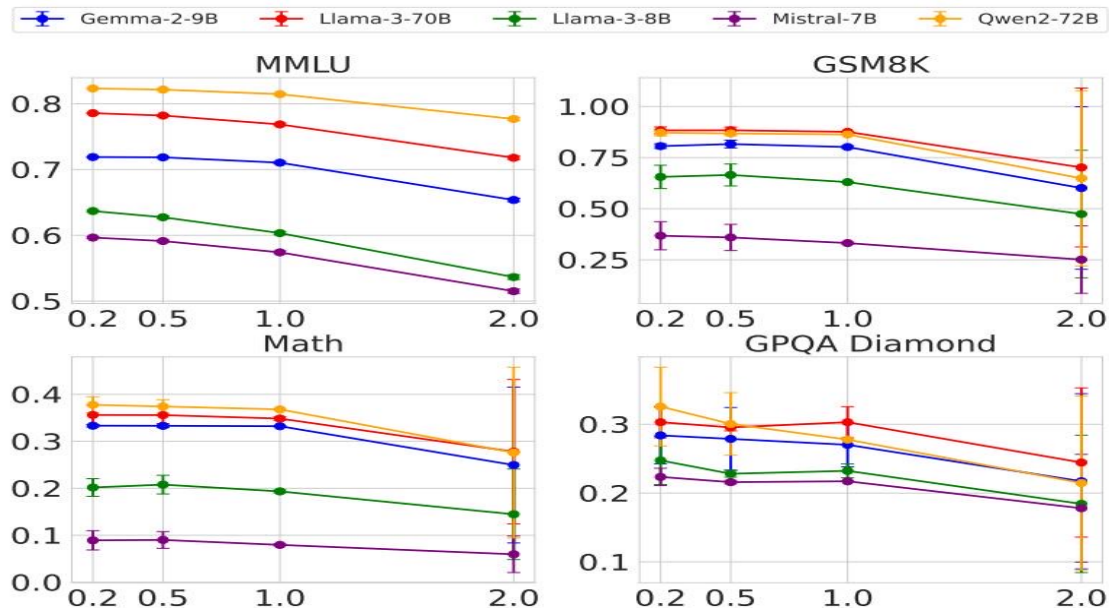


Figure 1: benchmark accuracy vs. temperature across five models and four tasks

What changes at higher temperature?

0.2 → 2.0

Experiments sweep temperatures 0.2, 0.5, 1.0, 2.0.

Degradation

At temperature 2.0, benchmark performance degrades across models.

Audit impact

Noise and narrowing gaps obscure the substitution signal.

Log Probability Fingerprinting Fails Due to Inference Nondeterminism

Model	BERT Acc	T5 Acc	GPT2 Acc	LLM2Vec Acc
Llama3-70B-Instruct-FP8	50.55	50.10	50.45	49.90
Llama3-70B-Instruct-INT8	51.60	49.90	51.30	50.25
Gemma2-9b-it-FP8	49.95	50.30	51.20	49.80
Gemma2-9b-it-INT8	49.00	49.70	51.65	49.55
Mistral-7b-v3-Instruct-FP8	50.55	49.75	48.70	48.75
Mistral-7b-v3-Instruct-INT8	49.70	50.75	50.50	51.15
Qwen2-72B-Instruct-FP8	50.05	50.25	48.75	49.80
Qwen2-72B-Instruct-INT8	50.75	50.55	49.45	50.20

Figure 5: same Llama-3-8B model, varying log probabilities across frameworks and GPUs

Reference idea

Compare API token log-probs against a local model fingerprint

Software

transformers 4.40 vs 4.50; vLLM 0.6.2 vs 0.8.2

Hardware

A100 vs H100 changes production inference behavior

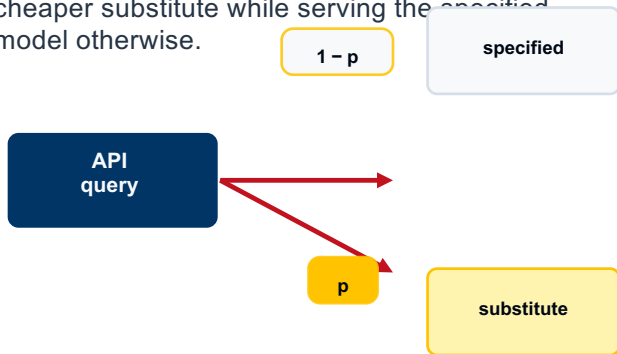
Consequence

Legitimate infrastructure drift can mimic substitution

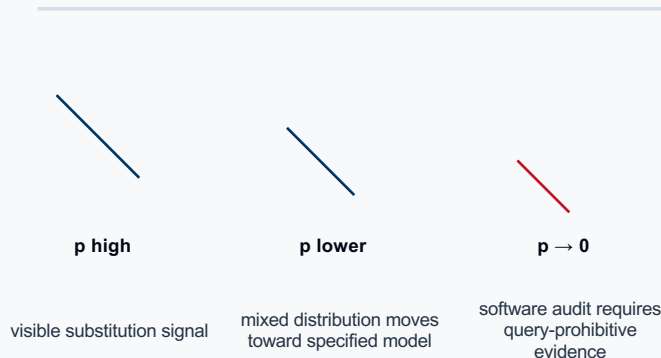
Randomized Routing at Low Rates Makes Statistical Detection Query-Prohibitive

Mixture attack

Provider routes only a fraction p of queries to the cheaper substitute while serving the specified model otherwise.



$$P_{\text{mix}}(y|x) = (1 - p) P_{\text{spec}}(y|x) + p P_{\text{sub}}(y|x)$$



As p approaches zero, P_{mix} converges to P_{spec} ; the statistical distance shrinks with the adversary's routing rate.

Software-Only Auditing Methods Are Fundamentally Unreliable

Method type	Weakness	Practical limitation
Text-based statistical tests	MMD and related output-distribution tests need impractically large query budgets and fail against subtle substitutions.	Low power when benchmark deltas are small or routing is randomized.
Log-probability methods	Token log-prob fingerprints vary across software frameworks and GPU hardware even for the same model.	Cannot separate legitimate infrastructure drift from actual substitution.
Adversarial countermeasures	Providers can use benchmark evasion or randomized substitution to hide behavior during audits.	Software-only audits become query-prohibitive and gameable.

Bottom line: software methods observe outputs or metadata, while the adversary controls serving policy, infrastructure, and benchmark routing.

Trusted Execution Environments Provide Provable Cryptographic Model Integrity

TEE attestation path

Model weights loaded



Inference code running



Cryptographic attestation

Proof concerns execution, not statistical resemblance.

Provable

Hardware-secured enclaves attest that specific weights and code executed.

Efficient

Practical today with only modest performance overhead.

Robust

Unaffected by benchmark evasion, randomized routing, or log-prob nondeterminism.

Deployment path: NVIDIA confidential computing on H100 GPUs makes TEE-backed inference a practical alternative; ZKPs remain computationally prohibitive for large models.

Actionable alternative: attest the runtime instead of guessing from outputs

Close the Verification Gap with Hardware Attestation

- 01 Model substitution is economically motivated and hard to observe through black-box APIs.
- 02 FP8 quantization barely affects many benchmark scores, weakening benchmark-based audits.
- 03 Production inference nondeterminism defeats log-probability fingerprinting.
- 04 Randomized substitution and benchmark evasion make statistical tests query-prohibitive.
- 05 TEEs provide the currently viable path to verifiable LLM API integrity.

Community action

Move LLM API integrity from probabilistic detection toward cryptographic verification of the model runtime.

Auditors should demand attestable inference claims, not just aggregate benchmark parity.